

Asterisk 18 Dialplan Applications – Complete Reference with Examples and WHY (43 Applications)

This document includes **43 commonly used Asterisk 18 dialplan applications** with: - Purpose - Full syntax - Parameter meanings - Possible values (type + examples) - Working dialplan example - **WHY each application is used in detail**

1. Answer

Purpose: Answer an incoming call

Syntax:

```
Answer([delay])
```

Parameters: | Name | Type | Meaning | Possible Values | |----|----|----|----| | delay | Integer | Delay before answering | Seconds (0–10) |

Example:

```
exten => s,1,Answer()
```

WHY: Explicitly accepts a call so audio, billing, recording, or subsequent dialplan apps can interact. Essential for IVR and AGI.

2. Playback

Purpose: Play an audio file

Syntax:

```
Playback(filename[,options])
```

Parameters: | filename | String | Sound file | `welcome` | | options | String | Playback options | `skip`, `noanswer` |

Example:

```
same => n,Playback(welcome)
```

WHY: Used for announcements or messages where input isn't needed.

3. Background

Purpose: Play audio while listening for DTMF input

Syntax:

```
Background(filename[,options])
```

Example:

```
same => n,Background(main-menu)
```

WHY: Used for IVR menus, allowing user selection during playback.

4. WaitExten

Purpose: Wait for user DTMF input

Syntax:

```
WaitExten([seconds])
```

Example:

```
same => n,WaitExten(10)
```

WHY: Pauses execution for user input after a prompt.

5. Read

Purpose: Capture DTMF input into a variable

Syntax:

```
Read(variable,filename[,maxdigits[,options[,timeout]])
```

Example:

```
same => n,Read(PIN,enter-pin,4)
```

WHY: Used for secure input like PIN codes or menu choices.

6. Set

Purpose: Set a channel variable

Syntax:

```
Set(variable=value)
```

Example:

```
same => n,Set(TRY=1)
```

WHY: Store runtime information for logic control or AGI scripts.

7. Dial

Purpose: Connect to another endpoint

Syntax:

```
Dial(technology/resource[,timeout[,options[,url]])
```

Example:

```
same => n,Dial(PJSIP/200,20)
```

WHY: Bridges calls to extensions, external numbers, or trunks.

8. Hangup

Purpose: Terminate the call

Syntax:

```
Hangup([causecode])
```

Example:

```
same => n, Hangup()
```

WHY: Ends calls cleanly with proper signaling.

9. Goto

Purpose: Jump to a specific context, extension, priority

Syntax:

```
Goto([context,]extension,priority)
```

Example:

```
same => n, Goto(ivr-menu,s,1)
```

WHY: Used to control call flow; essential for IVR and routing.

10. GotoIf

Purpose: Conditional branching

Syntax:

```
GotoIf(condition?true:false)
```

Example:

```
same => n,GotoIf($["${PIN}"="1234"]?success,1:fail,1)
```

WHY: Makes decisions based on variables or expressions.

11. While / EndWhile

Purpose: Loop execution

Syntax:

```
While(condition)
...
EndWhile
```

Example:

```
same => n,Set(I=1)
same => n,While($[${I}<=3])
    same => n,Playback(beep)
    same => n,Set(I=${I}+1)
EndWhile
```

WHY: For retries, repeated menus, or multiple attempts.

12. Macro

Purpose: Reusable logic (legacy)

Syntax:

```
Macro(name[,arg1[,arg2...]])
```

Example:

```
[macro-call-user]
exten => s,1,Dial(PJSIP/${ARG1},20)
```

WHY: Encapsulates reusable logic; deprecated in favor of Gosub.

13. Gosub / Return

Purpose: Modular reusable subroutine logic

Syntax:

```
Gosub(context,extension,priority)  
Return()
```

Example:

```
same => n,Gosub(utils,log,s,1)
```

WHY: Modern replacement for Macro; improves maintainability.

14. Progress

Purpose: Indicate call progress without answering

Syntax:

```
Progress()
```

Example:

```
same => n,Progress()
```

WHY: Used to play tones or announcements before billing.

15. Ringing

Purpose: Send ringing indication

Syntax:

```
Ringing()
```

Example:

```
same => n,Ringing()
```

WHY: Simulates ringing for the caller.

16. Busy

Purpose: Return busy signal

Syntax:

```
Busy([timeout])
```

Example:

```
same => n,Busy(10)
```

WHY: Indicates line busy or capacity exceeded.

17. Congestion

Purpose: Indicate network congestion

Syntax:

```
Congestion([timeout])
```

Example:

```
same => n,Congestion(5)
```

WHY: Used for controlled call failure instead of hangup.

18. Wait

Purpose: Pause execution

Syntax:

```
Wait(seconds)
```

Example:

```
same => n,Wait(2)
```

WHY: Timing control, wait for external processes.

19. MusicOnHold

Purpose: Play music on hold

Syntax:

```
MusicOnHold([class[,duration]])
```

Example:

```
same => n,MusicOnHold(default,10)
```

WHY: Keeps caller engaged while waiting.

20. StopMusicOnHold

Purpose: Stop MOH

Syntax:

```
StopMusicOnHold()
```

Example:

```
same => n,StopMusicOnHold()
```

WHY: Stops hold music once call resumes.

21. SendDTMF

Purpose: Send DTMF digits

Syntax:

```
SendDTMF(digits[,timeout])
```

Example:

```
same => n,SendDTMF(1234)
```

WHY: Used to interact with external IVR or systems requiring DTMF.

22. SayDigits

Purpose: Speak digits

Syntax:

```
SayDigits(digits)
```

Example:

```
same => n,SayDigits(1001)
```

WHY: Useful for reading account numbers or codes to caller.

23. SayNumber

Purpose: Speak a number

Syntax:

```
SayNumber(number[,gender])
```

Example:

```
same => n,SayNumber(250)
```

WHY: Used for verbalizing numeric information.

24. Verbose

Purpose: Log message to CLI

Syntax:

```
Verbose(level,message)
```

Example:

```
same => n,Verbose(2,Call reached IVR)
```

WHY: For debugging and logging call flow.

25. MixMonitor

Purpose: Record call audio

Syntax:

```
MixMonitor(filename[,options[,command]])
```

Example:

```
same => n,MixMonitor(/tmp/call.wav)
```

WHY: Used for compliance, quality monitoring, or dispute resolution.

26. StopMixMonitor

Purpose: Stop recording

Syntax:

```
StopMixMonitor()
```

Example:

```
same => n,StopMixMonitor()
```

WHY: Ends call recording cleanly.

27. Monitor

Purpose: Legacy recording

Syntax:

```
Monitor(type,filename[,options])
```

Example:

```
same => n,Monitor(wav,call001)
```

WHY: Used in older systems; replaced by MixMonitor.

28. ExecIfTime

Purpose: Execute app based on time

Syntax:

```
ExecIfTime(times,days,dates,months?app(args))
```

Example:

```
same => n,ExecIfTime(09:00-17:00,mon-fri,*,*?Playback(open))
```

WHY: Time-based routing for business hours or holidays.

29. GotoIfTime

Purpose: Time-based branching

Syntax:

```
GotoIfTime(times,days,dates,months?label)
```

Example:

```
same => n,GotoIfTime(09:00-17:00,mon-fri,*,*?open,s,1)
```

WHY: Route calls depending on time.

30. ChanSpy

Purpose: Listen to another channel

Syntax:

```
ChanSpy(channel[,options])
```

Example:

```
same => n,ChanSpy(PJSIP/200,q)
```

WHY: Supervisor monitoring, training.

31. Pickup

Purpose: Pickup a ringing call

Syntax:

```
Pickup([group])
```

Example:

```
exten => *8,1,Pickup()
```

WHY: Answer calls in a group without direct dialing.

32. Bridge

Purpose: Bridge two channels

Syntax:

```
Bridge(channel)
```

Example:

```
same => n,Bridge(PJSIP/300)
```

WHY: Advanced call control or supervisor intervention.

33. DB

Purpose: Read from AstDB

Syntax:

```
DB(family/key)
```

Example:

```
same => n,Set(BALANCE=${DB(users/100)})
```

WHY: Persistent storage for feature flags, user state.

34. DBPut

Purpose: Write to AstDB

Syntax:

```
DBPut(family/key=value)
```

Example:

```
same => n,DBPut(users/100=active)
```

WHY: Store or update data persistently.

35. DBDel

Purpose: Delete AstDB entry

Syntax:

```
DBDel(family/key)
```

Example:

```
same => n,DBDel(users/100)
```

WHY: Remove obsolete or temporary data.

36. SIPAddHeader

Purpose: Add SIP header (legacy)

Syntax:

```
SIPAddHeader(name: value)
```

Example:

```
same => n,SIPAddHeader(X-Test: 123)
```

WHY: For CRM integration or trunk tagging.

37. PJSIP_HEADER

Purpose: Manipulate SIP headers (PJSIP)

Syntax:

```
Set(PJSIP_HEADER(action,header)=value)
```

Example:

```
same => n,Set(PJSIP_HEADER(add,X-Source)=IVR)
```

WHY: Advanced SIP signaling manipulation.

38. TryExec

Purpose: Safely execute an application

Syntax:

```
TryExec(app(args))
```

Example:

```
same => n,TryExec(Dial(PJSIP/200,10))
```

WHY: Prevents dialplan crash if app fails.

39. ExecIf

Purpose: Conditional execution

Syntax:

```
ExecIf(condition?app(args))
```

Example:

```
same => n,ExecIf($[${BALANCE}>0]?Playback(ok))
```

WHY: Execute apps based on runtime conditions.

40. ReturnIf

Purpose: Conditional return from Gosub

Syntax:

```
ReturnIf(condition)
```

Example:

```
same => n,ReturnIf($[${AUTH}=0])
```

WHY: Exit subroutine based on logic.

41. MixMonitor

Purpose: Record the call

Syntax:

```
MixMonitor(filename[,options[,command]])
```

Example:

```
same => n,MixMonitor(/tmp/call.wav)
```

WHY: Call recording for compliance and QA.

42. StopMixMonitor

Purpose: Stop call recording

Syntax:


```
StopMixMonitor()
```

Example:

```
same => n,StopMixMonitor()
```

WHY: End recording cleanly.

43. Monitor

Purpose: Legacy recording application

Syntax:

```
Monitor(type,filename[,options])
```

Example:

```
same => n,Monitor(wav,call001)
```

WHY: Used in older systems; replaced by MixMonitor.

 This document now includes all 43 major Asterisk 18 dialplan applications with syntax, parameters, examples, and WHY explanations for each.